

Yahya Masri

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EDUCATION

George Mason University

Bachelor of Science, Computer Science (GPA: 4.0)

- **Coursework:** Data Structures, Formal Methods and Models, Introduction to Low-level Programming

May 2025 - Present

Northern Virginia Community College

Associate of Science, Computer Science (GPA: 3.875)

- **Coursework:** Linear Algebra

Aug 2023 - May 2025

EXPERIENCE

Lillup | *AI Software Engineer*

Jan 2026 - Present

- Built a **local RAG pipeline** that generates **source-cited** educational content using ChromaDB, LangChain, and GBNF-constrained generation via llama-cpp-python on quantized CPU models, resulting in consistently referenced material that reduces manual content curation effort
- Benchmarked three **on-device SLMs** across 20 queries each on parse reliability and hallucination rate, selecting Qwen3 0.6B at an average latency of **4.21s** with zero parse failures.
- Implemented **retrieval-stage grounding enforcement** through **cosine distance threshold filtering**, achieving **100% precision** and **96% accuracy** across 100 test queries, eliminating hallucinated output.

George Mason University | *Teaching Assistant*

Jan 2026 - Present

- Supported a **292-student** CS 262 course by promptly answering **conceptual and programming questions** on Piazza, which improved student satisfaction and helped maintain high assignment completion rates
- Explained **low-level programming concepts** clearly and concisely to help students understand course material and assignment requirements.

NSF Spatiotemporal Innovation Center, George Mason University | *Undergraduate Research Assistant*

Feb 2024 - Present

- Benchmarked **SLMS and SRLMs** on real-world Linux system logs, achieving **96% accuracy** with retrieval-augmented prompting, to evaluate deployability for real-time digital twin monitoring systems.
- Achieved **91% accuracy** on conflict-text classification by comparing **BERT and GPT** using hyperparameter tuning and prompting methods such as **RAG, RAG-ICL, and zero-shot** approaches.
- Optimized **context-based location-extraction** performance to **90%** through evaluation of open-source LLMs using various prompting methods and aligned evaluation methods.
- Reduced manual GIS processing time by **90%** through automated conflict-mapping workflows using a **multi-agent system (MAS)** integrated with QGIS, Geocoding API, and Streamlit.

PROJECTS

Glimpse | <https://tryglimpse.dev/>

- Co-developed an AI-powered product video startup that transforms a one-sentence product description into a cinematic 12-second video using a MAS with GPT-5.1 and Sora-2-Pro agents.
- Led product research and positioning, analyzing trends across recent YC batches to inform Glimpse's launch strategy focused on storytelling, differentiation, and waitlist validation.
- Implemented the core AI pipeline and backend, orchestrating the GPT-5.1 agents to Sora-2-Pro through a Python/OpenAI SDK stack to generate structured prompts and production-ready video outputs.
- Collaborated across a 4-person team to ship a functional end-to-end product from concept to demo, integrating AI, video generation, and product UX into a unified experience, using Git version control.

SoccerBot | <https://github.com/yahya3867/SoccerBot>

- Built a lightweight RAG pipeline through retrieval over curated league data using Python and CSV-based search, enabling context-aware LLM predictions.
- Enhanced prediction quality through structured prompt engineering, producing clearer, team-specific analyses with consistent reasoning.
- Created a UI through Streamlit using dropdown-driven team selection and RAG-backed retrieval logic, improving reliability with robust error handling for missing historical data.

SKILLS

- **Programming Languages:** Python, C, Java
- **ML & LLMs:** RAG, FAISS, LLM Orchestration, PyTorch, OpenAI API, Foundational Models, Vector Databases, Embeddings, Evaluation Pipelines
- **Data Processing & GIS:** Pandas, MongoDB, Neo4j, QGIS, Streamlit, ArcGIS Pro
- **Infrastructure & DevOps:** Git, Linux, OpenStack, Prometheus, Grafana
- **Tools & Frameworks:** VS Code, LM Studio, Unity

SELECTED PUBLICATIONS

- **Masri, Y.,** Ma, E., Wang, Z., Rogers, J. and Yang, C..Benchmarking Small Language Models and Small Reasoning Language Models on System Log Severity Classification.*arXiv preprint arXiv:2601.07790*
- **Masri, Y.,** Wang, Z., Srirenganathan Malarvizhi, A., Ahmed, S., Stover, T., Wong, D.W., Jiang, Y., Li, Y., Liu, Q., Bere, M. and Rothbart, D..Comparative Analysis of BERT and GPT for Classifying Crisis News with Sudan Conflict as an Example.*Algorithms, 18(7), p.420*
- Wang, Z., **Masri, Y.,** Malarvizhi, A.S., Stover, T., Ahmed, S., Wong, D., Jiang, Y., Li, Y., Bere, M., Rothbart, D. and Pfoser, D..Optimizing context-based location extraction by tuning open-source LLMs with RAG.*International Journal of Digital Earth, 18(1), p.2521786*